

Long-Term Behavior of Sea Turtle Populations Using Matrix Methods

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Course: APPM 3310 Matrix Methods

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Abstract:

This research examines the dynamics of the loggerhead sea turtle population for a longer period of time using the theories that have been introduced in the study of matrix methods, specifically APPM 3310. Specifically, this study focuses on determining how the survival and reproduction within each stage can impact the long term growth using the stage structured population model. In this research, we use the data from the study conducted by the Ecological Society of America regarding the seven stage loggerhead sea turtles population, which calculates the population size of surviving and reproducing sea turtles. The linear algebra needed to analyze this will be derived from the chapters in Applied Linear Algebra on topics concerning matrices and vectors, multiplication of matrices, determinants, eigenvalues and eigenvectors, eigenvector bases, and the diagonalization process. Firstly, we define the population vector and the seven-stage transition matrix. We then describe how multiplying the matrix allows us to project the state of the population in the future and how the dominant eigenvalue affects whether the population grows or declines in the long term. We discover that the dominant eigenvalue is about 0.945; therefore, the population of sea turtles will decline in the future. We also show that changes in later juveniles' and sub-adults' survivability have a significant impact on the growth factor. In the whole report, all numerical values in the sea turtle matrix are based on actual numbers from the references.

Attribution:

Mahi Chalicham worked on numerical analysis and matrix computations. Sanjitha Chakka worked on the background research and source data. Sanvitha Vallem worked on the eigenvalue analysis and long-term behavior sections. All group members contributed to writing, editing, and checking the mathematical work, numerical analysis, and building our code.

Introduction:

An example of a population that cannot be modeled using a simple one-stage growth model is the loggerhead sea turtle population. There are numerous life stages for these animals, including those of hatchlings, juveniles, subadults, and several breeding stages. For this reason, there are multiple parameters needed to describe a sea turtle population, which our stage-based growth model provides by dividing turtles into biologically meaningful groups and illustrating transitions between the stages.

As such, this paper focuses on the use of a suitable stage-based modeling method to explore the following ecological question relevant to a matrix approach: what impact do differences in reproductive and survival rates of various life stages have on population growth over time? The problem is a good fit for matrices because the sea turtle population in each point in time can be described by a vector, whereas transitions per year will be represented by a square matrix. All other information will follow through matrix operations.

Applied Linear Algebra uses the following chapters' content as its theoretical base: matrices and vectors (Section 1.2), matrix multiplication (Section 1.3), determinants (Section 1.9), eigenvalues and eigenvectors (Section 8.2), eigenvector basis (Section 8.3), and matrix diagonalization (Section 8.4). Sea turtle sources provide data and information about the seven life stages of the studied population, whereas our textbook helps us analyze the data.

Overall, it is established that the dominant eigenvalue of the baseline population matrix is smaller than one, which means that in the long run, this population will experience a decrease. This illustrates that any short-term population spikes may not ensure its stability in the long term. Thus, the transition matrix becomes the most crucial element in this particular problem.

The remainder of this paper is structured in the following manner. Within the Mathematical Formulation section, we will describe the state vector and construct a 7×7 projection matrix that depicts stage transition probabilities within the life cycle. Within the Numerical Results and Examples section, we will use matrix multiplication to predict short-term population dynamics and analyze eigenvalue methods and matrix diagonalization techniques for assessing long-term population dynamics. The Discussion and Conclusions section will then provide interpretations of our dominant eigenvalue and investigate the ecological consequences of improved survivorship within particular stages.

Mathematical Formulation:

In a stage-structured model, the population at time t is represented by a column vector. For the loggerhead model, there are seven stages, so the state vector is a vector in $\mathbb{R}^7$. If x_t denotes the population vector at year t , then each coordinate records the number of female turtles in one stage at that time. The source materials describe the seven stages as hatchlings, small juveniles, large juveniles, subadults, novice breeders, first-year remigrants, and mature breeders. Writing the population as a vector is important because it allows us to treat the biological system as a linear transformation.

Instead of writing each stage separately every year, we define one square matrix A , which describes how individuals move between stages from one year to the next, that sends the present stage x_t to the next stage x_{t+1} .

The matrix equation is

$$x_{t+1} = A x_t$$

This is the main formula in the model. The above equation is obtained from the matrix form of a discrete linear model based on our course vector-matrix multiplication concept. The idea is that after one time step of changes in the population, one gets the new population vector by simply multiplying the existing population vector with the matrix A . If we want to study how the population changes through many years, we repeat the process using A year after year.

$$x_{t+2} = A x_{t+1} = A(Ax_t) = A^2 x_t$$

Repeating this process shows a clear pattern: each additional year corresponds to multiplying by another copy of A. Continuing in this way, the population after k years is

$$x_{t+k} = A^k x_t$$

It can be seen from this formula that the effect of time on the development of the population can be considered by successive multiplication by the matrix A. Therefore, it is more appropriate to consider how the properties of A^k change when increasing k.

Also, all elements of this matrix have meaningful interpretations. The elements of the first line refer to the reproductive ability, which tells us about how many new members are born annually. The elements on the main diagonal tell us about the survival of turtles in their current phase and moving into next years in the same stage. Those below the main diagonal are responsible for surviving and moving into the next age category. All other elements are equal to zero since turtles cannot skip stages.

This matrix is taken from the literature on sea turtles' biology and our projects, while methods are known from class.

Construction of the Projection Matrix

The elements of the matrix rely on information about survival, length of stage, and reproductive rates taken from the source paper. For those stages where an individual spends several years, there should be considered probabilities of survival and remaining in the same class P_i , and the probability of survival and transitioning to the next stage G_i .

Suppose the survival rate per year in class is p_i and length of the period is d_i years. According to the source model, it is assumed that in this case there is stable distribution in the population in this stage. The corresponding proportions are equal to $1, p_i, p_i^2, \dots, p_i^{(d_i-1)}$. Summing these terms results in finding the sum of the elements inside the class and, correspondingly, expressions for both P_i and G_i .

$$P_i = ((1 - p_i^{(d_i-1)}) / (1 - p_i^{d_i})) p_i$$

$$G_i = (p_i^{d_i} (1 - p_i)) / (1 - p_i^{d_i})$$

Thus, in the case of Stage 2 (small juveniles), $p_2=0.7857$ and $d_2=7$ gives $P_2 \approx 0.7370$ and $G_2 \approx 0.0486$ which are the values used in the matrix. The same approach gives other parameters necessary for creating matrix.

As for one-year breeding stages (stages 5 and 6), no probabilities of staying in the stage should be considered since an individual can stay in this class only one year. It means that the probability of staying is zero and all those individuals who survived move to the next age stage. In the last class (Stage 7), mature breeders), an individual who survived remains in that stage.

After obtaining those probabilities, they are placed into the matrix A and then are utilized in further modeling.

Numerical Results and Examples:

Section 1. Matrix Multiplication and Population Transitions

The seven-stage loggerhead model begins by dividing the population into different life stages, each with its own survival and reproduction rates. The summarized stages as follows: Stage 1 is the hatchling stage and lasts one year. Stage 2 is the small juvenile stage and lasts seven years. Stage 3 is the large juvenile stage and lasts eight years. Stage 4 is the subadult stage and lasts six years. Stage 5 is the novice breeder stage and lasts one year. Stage 6 is the first-year remigrant stage and lasts one year. Stage 7 is the mature breeder stage and is the final breeding stage. Each stage lasts for a different amount of time, which affects how individuals move through the life cycle (Crouse, Crowder, and Caswell 1987; “Loggerhead Sea Turtle Project”).

Instead of treating each stage separately, these values are combined into a single 7×7 projection matrix A. This matrix summarizes all of the transitions between stages from one year to the next. The raw annual survival values given in there are 0.6747 for hatchlings, 0.7857 for small juveniles, 0.6758 for large juveniles, 0.7425 for subadults, 0.8091 for novice breeders, 0.8091 for first-year remigrants, and 0.8089 for mature breeders. The fecundity values are 0, 0, 0, 0, 127, 4, and 80. The sea turtle sources then convert these raw demographic values into the stage-to-stage matrix entries used in the final projection matrix (Crouse, Crowder, and Caswell 1987; “Loggerhead Sea Turtle Project”).

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 127 & 4 & 80 \\ 0.6747 & 0.7370 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.0486 & 0.6610 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.0147 & 0.6907 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.0518 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8091 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.8091 & 0.8089 \end{bmatrix}$$

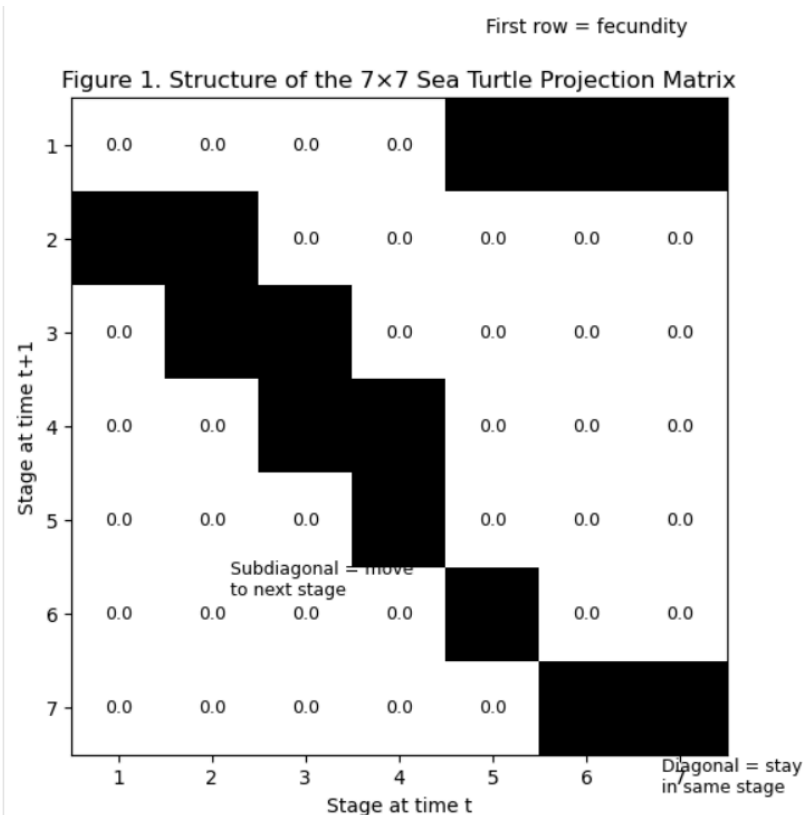


Figure 1. Structure of the 7×7 sea turtle projection matrix. The first row contains fecundity values, the diagonal entries represent remaining in the same stage, and the subdiagonal entries represent movement to the next stage (Crouse, Crowder, and Caswell 1987; “Loggerhead Sea Turtle Project”).

Figure 2. Summary of the Seven Stages in the Sea Turtle Model

Stage	Name	Role in model
1	Hatchlings	Early life stage
2	Small Juveniles	Growth and survival
3	Large Juveniles	Growth and survival
4	Subadults	Transition toward breeding
5	Novice Breeders	First breeding stage
6	1st-Year Remigrants	Post-breeding stage
7	Mature Breeders	Adult breeding stage

Figure 2. Summary of the seven stages used in the sea turtle population model. (Crouse, Crowder, and Caswell 1987; “Loggerhead Sea Turtle Project”).

The above matrix is the principal object of our analysis, and it is constructed according to the material from lectures. The first line of this matrix indicates that only the last three stages result in newly-born turtles. The lower lines describe survival and transitions between stages. For instance, 0.7370 in the second line and second column denotes survival of some small juveniles without changing their status to another one, whereas 0.0486 in the third line and second column means survival of small juveniles who become large juveniles after one more year of life.

The matrix may be read from columns as well. Each column describes what happens to turtles staying in a particular stage now. The first column implies that turtles staying in the hatchling stage do not have offspring, do not stay in the same stage, but move to the small juvenile stage with the probability 0.6747. The fifth column indicates that turtles staying in the novice breeder stage provide 127 hatchlings to the following generation, do not remain in the same stage, and move to the stage 6 with the survival probability 0.8091.

However, not all elements of the matrix belong to the same category of value. While some elements of the matrix reflect probabilities (values between 0 and 1), others indicate reproduction, which means that these values could be rather large. Therefore, a combination of different aspects of biological nature of objects is integrated in a single matrix. This is considered as one of the advantages of matrices for analysis since all transitions become represented by a single structure.

Figure 1 illustrates the structure of the projection matrix by presenting that the first row includes fecundity values, while diagonal elements indicate remaining in the same class and sub-diagonal elements indicate moving to another class.

Figure 2 represents the seven classes used in the projection matrix, their meaning, and biological interpretation. In particular, each class is characterized by its own biological interpretation, for instance, early classes are characterized by growth and survival and later classes are characterized by reproduction.

Section 2. Matrix Multiplication and Population Transitions

To make the model concrete, it helps to write out what multiplication by A actually does. Suppose the population at year t is

$$x_t = [x_1(t), x_2(t), x_3(t), x_4(t), x_5(t), x_6(t), x_7(t)]^T$$

The first coordinate represents the number of hatchlings in the next year. It is given by:

$$(Ax)_1 = 127x_5(t) + 4x_6(t) + 80x_7(t)$$

This means that only turtles in the breeding stages (stages 5, 6, and 7) produce new hatchlings. Each group contributes a different amount, based on the reproduction values in the matrix.

The second coordinate represents the number of small juveniles in the next year. It is given by:

$$(Ax)_2 = 0.6747x_1(t) + 0.7370x_2(t)$$

This shows that small juveniles come from two sources: hatchlings that survive and grow into stage 2, and small juveniles that survive and remain in the same stage. That interpretation is a direct example of matrix multiplication in an application setting.

Likewise,

The third coordinate is

$$(Ax)_t)_3 = 0.0486x_2(t) + 0.6610x_3(t)$$

The fourth coordinate is

$$(Ax)_t)_4 = 0.0147x_3(t) + 0.6907x_4(t)$$

The fifth, sixth, and seventh coordinates are

$$(Ax)_t)_5 = 0.0518x_4(t)$$

$$(Ax)_t)_6 = 0.8091x_5(t)$$

$$(Ax)_t)_7 = 0.8091x_6(t) + 0.8089x_7(t)$$

Overall, each coordinate is a weighted sum of the current population, where the weights come from the matrix. This shows how matrix multiplication naturally models survival, growth, and reproduction all at once.

$$x_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

This is our starting population vector, which represents the number of turtles in each stage at the beginning of the year.

$$Ax_0 = \begin{bmatrix} 211 \\ 1.4117 \\ 0.7096 \\ 0.7054 \\ 0.0518 \\ 0.8091 \\ 1.6180 \end{bmatrix} \quad A^2x_0 \approx \begin{bmatrix} 139.2550 \\ 143.4021 \\ 0.5376 \\ 0.4977 \\ 0.0365 \\ 0.0419 \\ 1.9634 \end{bmatrix}$$

This example shows how the population changes after one and two years, and gives a concrete picture of how the model works.

This illustrates how one multiplication by the projection matrix combines reproduction, survival, and stage transition into a new population vector. Repeating the process gives A^2x_0 , A^3x_0 , and so on, which is why powers of the matrix describe long-term behavior. After one year, the number of hatchlings increases significantly from 1 to 211 due to low reproduction rates relative to survival. After two years, the population continues to shift toward later stages, showing that early-stage turtles do not sustain the population long-term.

Writing out these coordinates helps show why the model is linear. Each stage in the next year is built from a combination of stages in the current year, using the values in the matrix. There are no nonlinear terms, so the model follows the rules of linear algebra.

We can repeat this same process to compute A^2x_0 , A^3x_0 , and so on. Even without calculating many steps, the matrix equation already shows how future populations are built from the current one. This leads to the main question: what happens to the population after many repeated applications of A ?

Section 3. Eigenvalue Analysis and Long-Term Stability

The key notion in long-term behavior studies is the notion of eigenvalues. For some scalar λ , a vector v is called an eigenvector, if

$$Av = \lambda v$$

This scalar is then an eigenvalue of A . Finding the eigenvalues involves solving a characteristic equation:

$$\det(A - \lambda I) = 0$$

In our case, the characteristic equation produces many eigenvalues, but one of them deserves special attention, because it is called the dominant eigenvalue. According to the sources provided, the dominant eigenvalue of the original matrix is approximately 0.945, which is less than 1, which guarantees long-term decline according to Crouse, Crowder, and Caswell (1987).

In case if the dominant eigenvalue were larger than 1, repeating multiplication by A would eventually lead to growth. With the eigenvalue being equal to 1, we would have the borderline case, where long-term population dynamics would remain unchanged. Since the dominant eigenvalue is less than 1, the size of the population eventually decreases each year, multiplying by roughly the same factor.

Within the framework of the provided model, the dominant eigenvalue is often referred to as the intrinsic growth factor. As was mentioned above, the value of the growth factor being around 0.945 means that, when the initial stage distribution effect fades, the population will behave just like if it were a geometric progression, with each year multiplying the number by 0.945, hence, leading to the decline of the population despite occasional deviations.

Another important aspect in long-term behavior prediction is finding eigenvectors. If the eigenvector is the vector corresponding to the dominant eigenvalue, then $A^k v = \lambda^k v$. It can be seen that the repeated use of a matrix to multiply the eigenvector does not alter its orientation, but merely rescales it. Therefore, the eigenvector corresponding to the dominant eigenvalue characterizes the eventual distribution of turtles across all stages.

Section 4. Diagonalization and Long-Term Behavior

Many concepts introduced in the previous sections can be better explained by means of diagonalization, which is an algebraic operation that transforms a matrix into a simpler equivalent matrix. In general terms, it allows for a more explicit representation of the effects of repeated matrix multiplications.

In this model, A is the matrix representing changes of population size per year, and x_t is the vector of population sizes at time t . The recurrence relation can be written as $x_{t+1} = Ax_t$, so multiplying matrix A repeatedly will give the population size k years after the initial observation. If A possesses a sufficient number of eigenvectors, it can be represented as:

$$A = S\Lambda S^{-1}$$

Here, S is a matrix formed from eigenvectors of the matrix, Λ is a diagonal matrix, and S^{-1} is the inverse matrix of S .

The second formula to find powers of A easily is:

$$A^k = S\Lambda^k S^{-1}$$

Such an identity is important since finding powers of matrices is usually difficult, while for the diagonal matrices it is a straightforward procedure. Suppose that Λ 's diagonal entries are eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_7$. Then diagonal entries of Λ^k are the k th power of $\lambda_1, \lambda_2, \dots, \lambda_7$. Thus, the term containing the largest eigenvalue will dominate after a while, as its k th power decreases the slowest.

This fact connects to section 3, where it was shown that the largest eigenvalue is around 0.945. Being smaller than 1, such a dominant eigenvalue means that, in the long run, the population will be shrinking since $0.945^k \rightarrow 0$ for $k \rightarrow \infty$. Therefore, in the end, only the behavior related to the dominant eigenvalue will matter.

Diagonalization explains why, in the long run, the initial population vector will play no significant role in future dynamics. An arbitrary vector x_0 can be represented as a linear combination of eigenvectors of A . As we multiply the vector by A every year, each vector component will scale according to the eigenvalue corresponding to the eigenvector.

The components associated with lower eigenvalues decay at a faster rate than the component associated with the highest eigenvalue, which dominates the other components in the long run. That is why the population tends to follow a consistent pattern of decline at a fairly stable rate in the long run, as we mentioned in Section 3.

It is because of this that the initial period of the population's growth may not show the same pattern as observed for an extended period. The population growth rate for the early periods would be impacted by the initial population structure, especially by the higher components of the lower eigenvalues.

Section 5. Impact of Stage-Specific Survival on Population Growth

Once we understand the meaning of the baseline matrix, we can make sense of what it tells us regarding survival. The information presented through the model states that survival of later juveniles and subadults has the largest effect on long-term growth. These turtles have survived their early hazards and are approaching adulthood, so their loss not only means a loss of survival but also their potential reproduction.

According to the sources provided, increasing the survival of these stages results in the dominant eigenvalue growing from approximately 0.945 to above 1. Therefore, the results change from long-term decrease to increase. Thus, not all stages play equal roles in the system, and some are far more important than others.

From the point of view of the matrix, this means that some matrix elements have a more significant effect on the long-term result than other elements. Additionally, it explains the mistake of focusing on hatchlings' numbers alone; despite their abundance, their low long-term survival can mean that the population will keep decreasing.

Discussion:

Through this project, one can understand the real-life problem solved using linear algebra techniques. Through the organization of sea turtle population into the matrix, matrix multiplication enabled us to determine the population growth in time. Rather than analyzing population stages individually, the matrix analysis helped us investigate all population stages together using a single model.

As for the most significant finding in the paper, the long-term behavior of the sea turtle population in the considered case is defined through its dominant eigenvalue. In our baseline model, the dominant eigenvalue equals to 0.945. Since it is smaller than 1, the population of the sea turtles is expected to decrease with time. Although during the first years there might be changes in the number of turtles, their population will continue to decline over time due to the repetitive multiplication process of the matrix.

The eigenvector corresponding to the dominant eigenvalue will define such a distribution of sea turtles in different stages. Thus, despite an overall decrease in the number of turtles, the distribution will converge in time.

The last interesting point is the difference in influence on the dominant eigenvalue of different life stages on the long-run growth of populations. As it follows from the results, the most essential influence is made by juvenile and subadult stages. If survival rates increase in these stages, then the dominant eigenvalue increases, and it becomes more than 1, which means growth of population. It means that it is more reasonable for improving survival of sea turtles rather than just reproducing them.

In addition, the example also illustrates the difference between short-run and long-run behavior of a population. In the first case, behavior depends on the initial conditions of a system and can change during time; in the second case, behavior is determined solely by the structure of matrix. That is why studying powers of the matrix gives much better understanding.

All things considered, the current example showed how to use matrix theory for explaining real-world problems.

Comparison of Survival Scenarios:

To determine the impact of survival rates on the growth rate of the population, the baseline model is compared with another modified model in which there is an increase in the survival rate in later juvenile classes. The comparison is made by finding out the dominant eigenvalue for each of the two models, because the dominant eigenvalue is the one that determines the fate of the population whether growing or declining.

The dominant eigenvalue in the baseline model is around 0.945 which means that the population will decline over time because it is less than 1. In the modified model, there is an increase in the dominant eigenvalue to around 1.02 meaning that the population grows because it is more than 1.

This analysis indicates that different life history stages do not affect growth in the same way. Specifically, increases in survival in the late juvenile/subadult stages have a far greater impact than improvements at earlier life history stages. This is due to the fact that individuals in those stages are much closer to being breeders, and thus improvements in their survival increase numbers and breeding.

In summary, this graph highlights the influence of survival rates on the dominant eigenvalue.

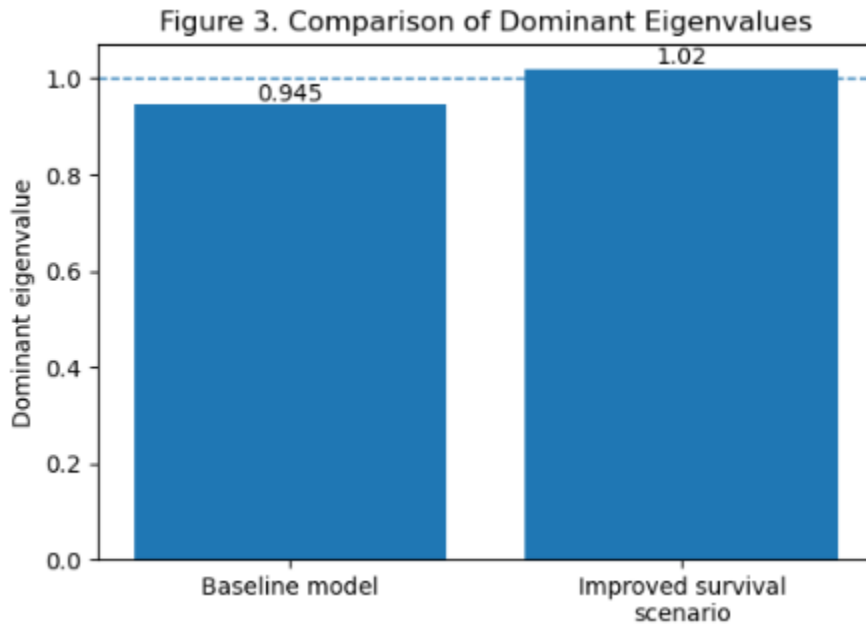


Figure 3. Comparison of the dominant eigenvalue for the baseline and modified survival scenarios.

Assumptions of the Model

Like any other mathematical model, there are some assumptions involved in this sea turtle model. To start, the entries in the matrix remain constant through time. That means the survival rate as well as the reproductive rate will be the same for each year regardless of the changing environmental conditions.

Another assumption is that the model is a linear one because every year's population is considered to be a linear combination of its previous year. In such a way, there will be no considerations such as overpopulation and competition included.

Thirdly, the model considers a stage-specific approach instead of an age-specific one. It means that the age of the turtles is unknown and therefore we consider developmental stages instead.

Fourthly, in this model only the females are considered as reproducers and thus their values are used to determine fecundity entries which also come from the source model.

Conclusion

Matrix techniques were utilized in this project to examine the asymptotic dynamics of a stage-structured population of loggerhead sea turtles. The population was presented using a vector in R^7 , and annual dynamics were formulated as a 7×7 projection matrix. Matrix operations and powers of matrices were employed to study the dynamics of the system, while eigenvectors and eigenvalues were used to investigate its long-term behavior.

The main findings of the work include the determination that the matrix associated with the baseline scenario possesses a dominant eigenvalue of about 0.945. Since this value is smaller than one, the conclusion can be made that the population will experience an asymptotic decline. The results of the analysis are independent of the initial population state. This is explained by the diagonal form of the matrix since the eigenvalue governs the system dynamics in repeated applications of the matrix.

This study further proves that survival in different life stages has varying effects on the long-term growth of the population under study. More specifically, it proves that the survival rate at the late juvenile and sub-adult stages have the highest effect on the dominant eigenvalue. Whenever survival rates at those two stages are improved, the dominant eigenvalue will exceed one and consequently cause growth in population.

Generally, from this study, the use of matrix analysis can effectively be used to describe the behavior of actual phenomena. Through the use of vectors, matrices, eigenvalues, and matrix diagonalization, among others, we have managed to effectively describe the trend of the population.

The linear model that we used currently does provide insight on the population dynamics of the loggerhead sea turtle, but there are definitely ways that we can improve in the future. First off, our linear model can be made more realistic by converting it to a stochastic model, which would mean having our projection matrix parameters change every year based on factors such as unpredictable shifts in temperature due to climate change. Second, an official elasticity analysis using both our left and right eigenvectors will help us understand which life stage transition is impacted the most by conservation efforts.

References

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3. "Loggerhead Sea Turtle Project." Math Insight. https://mathinsight.org/assess/math2241/sea_turtle_project?
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Appendix

Figure 1: Sea Turtle Projection Matrix Structure Code

```
import matplotlib.pyplot as plt
import numpy as np

# this is the original 7x7 sea turtle matrix
A = np.array([
    [0, 0, 0, 0, 127, 4, 80],
    [0.6747, 0.7370, 0, 0, 0, 0, 0],
    [0, 0.0486, 0.6610, 0, 0, 0, 0],
    [0, 0, 0.0147, 0.6907, 0, 0, 0],
    [0, 0, 0, 0.0518, 0, 0, 0],
    [0, 0, 0, 0, 0.8091, 0, 0],
    [0, 0, 0, 0, 0, 0.8091, 0.8089]
])

# make the graph
fig, ax = plt.subplots(figsize=(7, 6))

# color the nonzero entries so the structure is easier to see
ax.imshow(A != 0, cmap="Greys", interpolation="none")

# put each number inside its spot in the matrix
for i in range(7):
    for j in range(7):
        ax.text(j, i, A[i, j], ha='center', va='center', fontsize=9)

# set the stage numbers on both axes
ax.set_xticks(range(7))
ax.set_yticks(range(7))
ax.set_xticklabels([1, 2, 3, 4, 5, 6, 7])
ax.set_yticklabels([1, 2, 3, 4, 5, 6, 7])

# label axes
ax.set_xlabel("Stage at time t")
ax.set_ylabel("Stage at time t+1")

# title
ax.set_title("Figure 1. Structure of the 7x7 Sea Turtle Projection Matrix")

# add simple labels to explain the matrix
ax.text(5, -1.2, "First row = fecundity", ha='center', fontsize=10)
ax.text(5.8, 6.9, "Diagonal = stay\nin same stage", ha='left', fontsize=9)
ax.text(1.2, 4.8, "Subdiagonal = move\nto next stage", ha='left', fontsize=9)

# make spacing look better
plt.tight_layout()

# show graph
plt.show()
```

Figure 2: Summary of the Seven Stages Code

```
import matplotlib.pyplot as plt

# this is the stage information for the sea turtle model
data = [
    ["1", "Hatchlings", "Early life stage"],
    ["2", "Small Juveniles", "Growth and survival"],
    ["3", "Large Juveniles", "Growth and survival"],
    ["4", "Subadults", "Transition toward breeding"],
    ["5", "Novice Breeders", "First breeding stage"],
    ["6", "1st-Year Remigrants", "Post-breeding stage"],
    ["7", "Mature Breeders", "Adult breeding stage"]
]

# column names for the table
columns = ["Stage", "Name", "Role in model"]

# make the figure
fig, ax = plt.subplots(figsize=(9, 4))

# turn off the axes since this is just a table
ax.axis("off")

# create the table
table = ax.table(
    cellText=data,
    colLabels=columns,
    loc="center",
    cellLoc="center"
)

# keep the font size fixed
table.auto_set_font_size(False)

# set the table font size
table.set_fontsize(10)

# make the table a little bigger so it is easier to read
table.scale(1.2, 1.5)

# add the title
plt.title("Figure 2. Summary of the Seven Stages in the Sea Turtle Model", pad=12)

# show the table
plt.show()
```

Figure 3: Comparison of Dominant Eigenvalues Code

```
import matplotlib.pyplot as plt

# Labels for the two scenarios
labels = ["Baseline model", "Improved survival\nscenario"]

# dominant eigenvalues for each case
values = [0.945, 1.02]

# make the bar graph
plt.figure(figsize=(6, 4))
plt.bar(labels, values)

# add a dashed line at 1 to show the growth/decline cutoff
plt.axhline(1.0, linestyle="--", linewidth=1)

# put the values above each bar
for i, v in enumerate(values):
    plt.text(i, v + 0.01, str(v), ha='center')

# label the y-axis
plt.ylabel("Dominant eigenvalue")

# title
plt.title("Figure 3. Comparison of Dominant Eigenvalues")

# show the graph
plt.show()
```

AI Disclaimer: We used AI to check for grammar and improve wording clarity.